

DATA INTAKE

Raw scRNA-seq + TCR-seq

PREPRO

100,067 cells, 4 patients, 20k+ genes

Quality Control and Normalization

FE

Ambient RNA removal, doublet filtering, log-normalization

Parallel Feature Engineering

INTEGRATION

Transcriptomics Branch

HVG selection (1,500) and PCA (50 PCs)

TCR Branch

k-mer and physicochemical descriptors

Integrated Multimodal Matrix

4 feature families (29–429 features)

